Project Proposal

Regression

Background and Response

For the regression portion of this project I'll be looking at **official building electricity usage records** from Seattle. Non-residental and multi family homes are required to report energy usage to the city of Seattle, and this dataset contains 9 years of data from a ton of buildings. The full dataset contains over 46,000 records and over 40 columns of information for each building. I believe the value of this kind of analysis is apparent. Understanding what buildings are more or less efficient than others can help us understand what opportunities we have for improvement. Energy efficiency effects almost every single aspect of our lives. Gains there can put less strain on the power grid and lower usage of fossil fuels that provide this electricity.

For the response, I'll be using a variable called **EUI(kBtu/sf)**. This gives us a measure of electrical intensity per square foot, which allows us to compare the efficiency of buildings of various sizes. This is far better than pure electrical usage, where all I would determine is that bigger properties use more electricity. As well, there is a version of EUI that is weather normalized in this dataset, **EUIWN**, so it controls for bizarre weather that could impact electricity usage and throw off values.

Variables

I have a lot to choose from in this dataset. Currently, my interest is on a handful of columns. Below is a table of the response and all of my current candidates.

Variable	Purpose	Data Type	Description
EUIWN	Response	Numeric	Weather normalized electricity usage per square foot. Units: (kBtu/sf)
DataYear	Predictor	Integer	Year data was collected, ranges from 2015 - 2025
BuildingType	Predictor	Categorical	7 categories. Multifamily, hospital, non-residental, etc.
Neighborhood	Predictor	Categorical	14 categories. Describes general location of the building.
YearBuilt	Predictor	Integer	Age of the building.
NumberofBuildings	Predictor	Integer	Number of buildings on the property.
NumberofFloors	Predictor	Integer	Number of floors

These are all just candidate predictors so I'll have to determine which ones I want to include through analysis later on.

Preprocessing and splitting

There isn't a lot to do to pre-process this data as it is quite well behaved. `Neighborhood` has enough categories that I may need to group some together to keep things feasible. `DataYear` will both function as a predictor and as a splitting variable. So I'll likely keep 2025 as the test set. I need to also consider how I will handle the same building showing up multiple times over the years. Basically all of them do, some every single year. One approach is to randomly select a single row to keep for each building. This prevents data leakage and makes interpretation a lot simpler. An alternative solution is to keep one year for training and another for testing, so it becomes more about predicting the next years usage. I personally prefer the former approach as we can also get information on if electrical efficiency is going up or down over time from its coefficient.

Modeling and Evaluation

For the models I'll be sticking to the required ones on the rubrick. This is a straightforward dataset, so traditional linear regression, a point prediction NN and a probabilistic neural network should all provide some interesting results.

For metrics, I'll be sticking to the rubrick. RMSE, MAE, gaussian negative log likelihood, prediction interval coverage and average interval width will all be evaluated. I think that's all pretty thorough.

Proposed Investigation

I have a clear investigation I want to do on this dataset. I mentioned potentially taking only one row per building to prevent data leakage. What happens to the models if I don't do that? How much, if at all, do the model attributes and predictions change? For this experiment, I would keep a set of buildings as a completely separate hold out set with one row per building. In theory, the models with duplicate buildings may end up overfitting and perform worse on buildings it hasn't been exposed to. Will that end up being true?

For this we can compare the equivalent models with a simple paired hypothesis test. The null hypothesis will be that the point estimates for those locations are the same across both cases. We can do the same for interval widths as well. I'm curious to see what the results of this would be.

Classification

Background and Response

For the classification portion of this project I will be examining the **mushroom** dataset from the **Audobon Society Field Guide**, sourced from the **UC Irvine Machine Learning Repository**. This dataset contains a ton of categorical information about 8124 mushrooms with one observation per mushroom. The response of interest is whether a given mushroom is poisonous or not. As such, this

topic follows the classic style of fraud detection problems. Though the topic of mushroom edibility doesn't matter a lot on its own, the top of things being safe to consume has gotten far more relevance recently. Lately we have seen a staggering amount of FDA food recalls, and the current administration is consistency gutting and declawing various agencies that ensure what we put in our bodies is safe. So I feel this is topical in its own way as we can imagine various experts using similar models to detect dangerous food and drugs. I like using mushrooms here as it's tangentially related but not nearly as depressing.

Variables

Alongside the response are 22 other features, all categorical in nature. These describe aspects of the mushroom such as the coloring of the gills, whether there is any bruising on the mushroom, and the habitat it was found in. This aspect of the dataset I find extremely interesting as categorical columns have to be treated with an additional level of care to numeric ones. A categorical column may be a multi-level factor, an ordinal column, or one that can be treated as a simple boolean. This dataset has all of the above, so I have to be careful during the modeling setup to ensure I'm appropriately handling the final columns I choose.

Preprocessing and Splitting

This also leads into the topic of splitting the data. The response of this data is quite balanced, with 48% of the rows being poisonous and 52% being edible. This cannot be said for many of the categorical columns, with some particularly rare levels of them only showing up about 4 times. Almost every single column is extremely unbalanced. This leads to a risk of levels only being present in one of the splits, and with some categories being so infrequent not even stratification is a perfect solution. I will likely need to combine certain infrequent categories into an "other" level or omit them entirely to prevent breaking issues.

My plan to tackle this is to go through each multi level candidate variable and look for any obvious relationships with the response. For anything that seems promising, I'll decide on a case-by-case basis what feature engineering will be most appropriate. The simplest case are variables with only 2 levels, as they can be easily converted into boolean columns. The only ordinal column present in this dataset, `ring_number`, seems to be a simple integer column that was saved as text which is an easy fix. Lastly, any multi level categorical columns will use dummy variables. As for missing values, only the column `stalk_root` has any. Over a quarter of the rows $\left(\frac{2480}{8124}\right)$ are missing data for this column. As such, it will not be a candidate for this project. It's possible I will have to drop some rows depending on my columns of choice. Take `cap_surface`. It has 4 categories, and 3 are very balanced each having about 2 to 3 thousand rows. One category though, `grooves`, only has 4 rows. There is no reasonable way to include that one in the other categories, so I may need to simply drop those 4 rows entirely if I want to use that variable.

Modeling and Evaluation

For modeling, I'll be sticking with the required models from section 5.1. So that is logistic regression and a neural network classifier. For metrics I'll be focusing primarily on precision, recall and f-1 score. Recall is huge here in particular as this is a fraud-detection problem. False positives are less of a concern as there isn't a consequence to not eating a mushroom. False negatives however are extremely bad so we want to minimize the number of those.